
Let It Cook: A Comparative Study of Chemical Encoders for Synthesis Prediction

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Abstract

Current generative models for molecule synthesis struggle to reconstruct complex, real-world therapeutics, dropping significantly in performance on out-of-distribution datasets like ChEMBL. This limitation primarily stems from their reliance on scratch-trained encoders that lack an inherent understanding of rich chemical topology. To address this bottleneck, we replace the SynFormer encoder with frozen Pre-trained Chemical Language Models (PLMs). Because a direct encoder substitution fails due to a severe representation space mismatch, we introduce a trainable non-linear bridge layer to geometrically align the frozen PLM embeddings with the fine-tuned decoder. Evaluated on a controlled 500,000-sample training budget, our MolFormer-integrated architecture achieves a 100% success rate and a mean Tanimoto similarity of 0.176, outperforming both the SynFormer baseline (0.151) and ChemBERTa-integrated variant (0.155) at the same data scale. These results confirm that leveraging frozen PLM priors for explicit synthesis generation is architecturally sound and that performance is bounded primarily by training scale rather than model design.

1. Introduction

The discovery and design of novel therapeutics is a complex, resource-intensive process. Generative artificial intelligence has made significant strides in designing new molecular structures, but many early models focused solely on the final structure without considering whether the molecule could actually be synthesized in a laboratory. This gap has

shifted the field toward *synthesis-aware* generative models, which aim to produce not only a target molecule but also a valid, step-by-step reaction pathway using commercially available building blocks and established reaction templates.

Despite this progress, current state-of-the-art models still fail to generalize to real-world data. SynFormer (Gao et al., 2025), a 229-million parameter encoder-decoder Transformer for synthesis prediction, achieves a 66% exact reconstruction rate on simulated synthetic datasets. However, its performance collapses to approximately 20% on the ChEMBL (Zdrazil et al., 2023) dataset of bioactive, real-world drugs. This severe degradation reveals a fundamental architectural limitation: SynFormer’s encoder is trained from scratch solely on synthesis tokens, and thus lacks any pre-existing understanding of general chemical structure or molecular topology.

We propose replacing this scratch-trained encoder with dedicated Pre-trained Chemical Language Models (PLMs), which have already internalized rich representations of functional groups, substructures, and molecular topology from billions of molecules. We investigate two PLMs: MolFormer-XL (Ross et al., 2022), trained on 1.1 billion molecules using linear attention to capture long-range structural dependencies, and ChemBERTa (Chithrananda et al., 2020), a RoBERTa-style model trained on 77 million molecules with standard bidirectional attention. By freezing the PLM weights and introducing a trainable non-linear bridge layer, we isolate the fine-tuning to the decoder components alone, preserving the pre-trained chemical knowledge while teaching the decoder to translate rich PLM embeddings into valid synthetic instructions.

A key finding of our work is that direct encoder substitution is insufficient: MolFormer and SynFormer’s original encoder operate in fundamentally incompatible representation spaces, despite producing identically-sized output tensors. Our systematic investigation of this failure, and the design of the non-linear bridge layer as a solution, constitutes a core contribution. Our controlled three-way comparison at a fixed 500k-sample budget provides the first systematic evaluation of whether PLM integration provides a structural advantage over a scratch-trained baseline, and whether pre-training data scale or attention mechanism design matters

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more.

2. Related Work

Synthesis-Aware Generation. Early generative models such as Junction Tree VAE (Jin et al., 2018) and Equivariant Diffusion Models (Hoogeboom et al., 2022) demonstrated strong chemical space exploration but did not constrain outputs to synthesizable structures. To address this, models like ChemProjector (Luo et al., 2025) couple a Graph Neural Network (GNN) encoder with a Transformer decoder to project molecules into synthesizable spaces, achieving a 13% reconstruction rate on ChEMBL. Policy Gradient for Forward Synthesis (PGFS) (Gottipati et al., 2020) frames synthesis as a Markov Decision Process enforcing synthesizability at each step. SynLlama (Sun et al., 2025) fine-tunes Llama-3 on synthesis instruction data, treating chemistry as a pure language task, but risks hallucinating chemically invalid reactions due to the absence of hard chemical constraints and requires immense computational resources. SynFormer (Gao et al., 2025), which our work builds upon, avoids these issues by constraining the decoder to operate directly over a validated catalog of building blocks and reaction templates.

Chemical Language Models. MolFormer-XL (Ross et al., 2022) uses linear attention to scale to 1.1 billion SMILES strings, capturing long-range molecular dependencies efficiently. ChemBERTa (Chithrananda et al., 2020) employs standard RoBERTa-style bidirectional attention on 77 million molecules, producing strong embeddings for property prediction. MegaMolBART (NVIDIA, 2026) uses a sequence-to-sequence architecture for chemical understanding, while Uni-Mol (Zhou et al., 2023) extends the paradigm to universal 3D representations. The key distinction between these models is their training objective: all are trained on self-supervised molecular tasks (masked language modeling or similar), not on synthesis prediction.

The Integration Gap. Despite the strong molecular representations learned by these PLMs, their application in explicit step-by-step synthesis generation pipelines remains underexplored. SynFormer still relies on a scratch-trained encoder, isolating it from the rich structural priors available in MolFormer and ChemBERTa. Our work is the first to systematically investigate how to bridge this gap, quantifying the failure of naive substitution and introducing a principled adapter architecture that enables PLM integration while preserving decoder synthesis knowledge.

3. Methodology

3.1. SynFormer Architecture Overview

SynFormer employs an encoder-decoder Transformer designed explicitly for synthesis-aware molecule generation.

Table 1 provides a comprehensive breakdown of its architectural hyperparameters. Unlike standard autoregressive language models, the SynFormer decoder does not generate output token-by-token with the previous output as context. Instead, it maintains an explicit *synthesis stack* representing the current state of the reaction pathway. At each generation step, the entire current stack (all building blocks and reactions chosen so far) is re-fed into the decoder simultaneously, and the decoder predicts only the next action. This stack-based approach is critical: it ensures that the decoder always has global context about the synthesis state and can reason about chemical compatibility across the entire route.

To handle the vast catalog of 223,244 commercially available building blocks, SynFormer uses a continuous diffusion head attached to the decoder output. Rather than selecting a building block from a discrete vocabulary, the model predicts a continuous Morgan fingerprint embedding, which is then mapped to the nearest real building block via nearest-neighbor search in fingerprint space. This design guarantees that every output corresponds to a physically purchasable molecule.

Table 1. SynFormer Model Architecture and Training Details

Component	Details
Embedding Dimension	768
Positional Encoding	Sinusoidal
Encoder Layers	6
Decoder Layers	10
Attention Heads	16 per layer
Q/K/V Dimension	256 per head
FFN Dimension	4096
Token Types	[START], [END], [RXN], [BB]
Building Blocks	223,244 molecules
Fingerprint Size	2048 bits
Diffusion Steps (T)	100
Original Training Data	~742M samples
Original Optimizer	AdamW, LR 3×10^{-4}
Original Batch Size	1024
Original Training Time	~6 days (8× A100 GPUs)

The encoder’s role is simple but critical: it processes the input SMILES string and produces a fixed set of contextual embeddings $\mathbf{c} \in \mathbb{R}^{L \times 768}$, where L is the sequence length. These embeddings serve as the cross-attention memory for the decoder at every decoding step — the encoder is used exactly once, but its output guides the entire generation. This means the quality and compatibility of the encoder’s representation space directly determines how well the decoder can generate synthesis routes.

3.2. Proposed Architecture

Figure 1 illustrates the complete proposed pipeline. The original SynFormer encoder is replaced by a frozen PLM, followed by a trainable bridge layer that projects the PLM’s embeddings into the synthesis-specific representation space expected by the decoder.

3.3. The Representation Gap: Plug-and-Play Failure

Our initial hypothesis was that directly substituting the scratch-trained SynFormer encoder with MolFormer-XL would improve generalization to ChEMBL. However, zero-shot replacement produced no valid synthesis routes whatsoever.

Although both models output tensors of identical dimensionality ($[\text{batch}, \text{seq}, 768]$), they operate in fundamentally different representation spaces. MolFormer’s embeddings are optimized for generalized chemical similarity through masked language modeling on 1.1 billion molecules. In contrast, the SynFormer decoder’s cross-attention was co-trained with its original character-level encoder to expect synthesis-aware representations — embeddings that specifically encode which reactions and building blocks are likely to appear in a valid synthesis route for a given molecule. These two spaces are not linearly alignable: the geometric structure that clusters molecules by chemical similarity (MolFormer) is categorically different from the structure that clusters molecules by synthesizability patterns (original SynFormer encoder).

Concretely, passing MolFormer embeddings into the SynFormer decoder resulted in meaningless cross-attention signals. The token head predicted token types (REACTANT, REACTION) with near-uniform probability, the fingerprint diffusion head generated fingerprints with no structural grounding, and the retrieved building blocks were chemically incompatible with the predicted reactions. RDKit rejected nearly all synthesis routes, and the beam search states were aborted. The result was a 0% success rate for all 1,000 test molecules.

3.4. Bridge Layer Design

To reconcile these disparate representation spaces, we introduce a trainable non-linear bridge layer placed between the frozen PLM encoder and the SynFormer decoder. The bridge must accomplish two things simultaneously: (1) geometrically restructure MolFormer’s chemical similarity space into the decoder’s synthesis-specific space, and (2) preserve as much of the PLM’s chemical knowledge as possible in the transformed representation.

We first evaluated a linear bridge ($\text{Linear}(768, 768) + \text{LayerNorm}$), which can only rotate, scale, and shift the

representation space. This produced 0% success rate at both 10k and 500k training samples, confirming that a purely linear transformation is fundamentally insufficient to bridge the geometric gap between these two spaces.

Our final bridge architecture is a two-layer non-linear MLP:

$$\mathbf{c} = \text{LayerNorm}(\mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{h}_{\text{PLM}} + \mathbf{b}_1) + \mathbf{b}_2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{1536 \times 768}$, $\mathbf{W}_2 \in \mathbb{R}^{768 \times 1536}$. The intermediate expansion to 1536 dimensions creates a workspace in which the GELU activation can selectively suppress, amplify, and recombine features. This enables a true geometric restructuring — moving clusters of molecules from similarity-based positions to synthesizability-based positions — rather than a simple affine transformation.

3.5. Vocabulary and Tokenization Differences

The representation gap is further compounded by a fundamental mismatch in tokenization. SynFormer uses a character-level SMILES tokenizer with a vocabulary of approximately 150 tokens, where each token corresponds directly to a single atom symbol, bond, or SMILES punctuation character. MolFormer uses Byte-Pair Encoding (BPE) with 2,362 tokens, where frequent character combinations are merged into single tokens (e.g., CC, CN, c1c). ChemBERTa also uses BPE with approximately 600 tokens.

This discrepancy has a concrete structural consequence: for the same SMILES string, MolFormer produces a *shorter* sequence than the original encoder, while ChemBERTa produces sequences of intermediate length. The decoder’s cross-attention patterns were calibrated for a particular sequence length distribution. Furthermore, the special tokens [CLS] and [SEP] prepended and appended by the PLM tokenizers have no counterpart in the original character-level tokenizer, further shifting the positional structure that the decoder’s cross-attention mechanism has learned to expect.

3.6. Training Strategy and Computational Details

The PLM encoder weights are completely frozen throughout training. We jointly fine-tune the newly initialized bridge layer, the pre-trained SynFormer decoder, and all three prediction heads. To prevent catastrophic forgetting of the decoder’s pre-trained synthesis logic, we use a conservative learning rate of 3×10^{-5} — an order of magnitude lower than the original SynFormer training rate of 3×10^{-4} . We use AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01, and a ReduceLROnPlateau scheduler with factor 0.8 and patience 10 validation rounds.

Training data is generated on-the-fly from 223,244 commercially available Enamine building blocks and 115 curated reaction templates, without requiring any pre-collected dataset. At each step, a random valid synthesis route is sam-

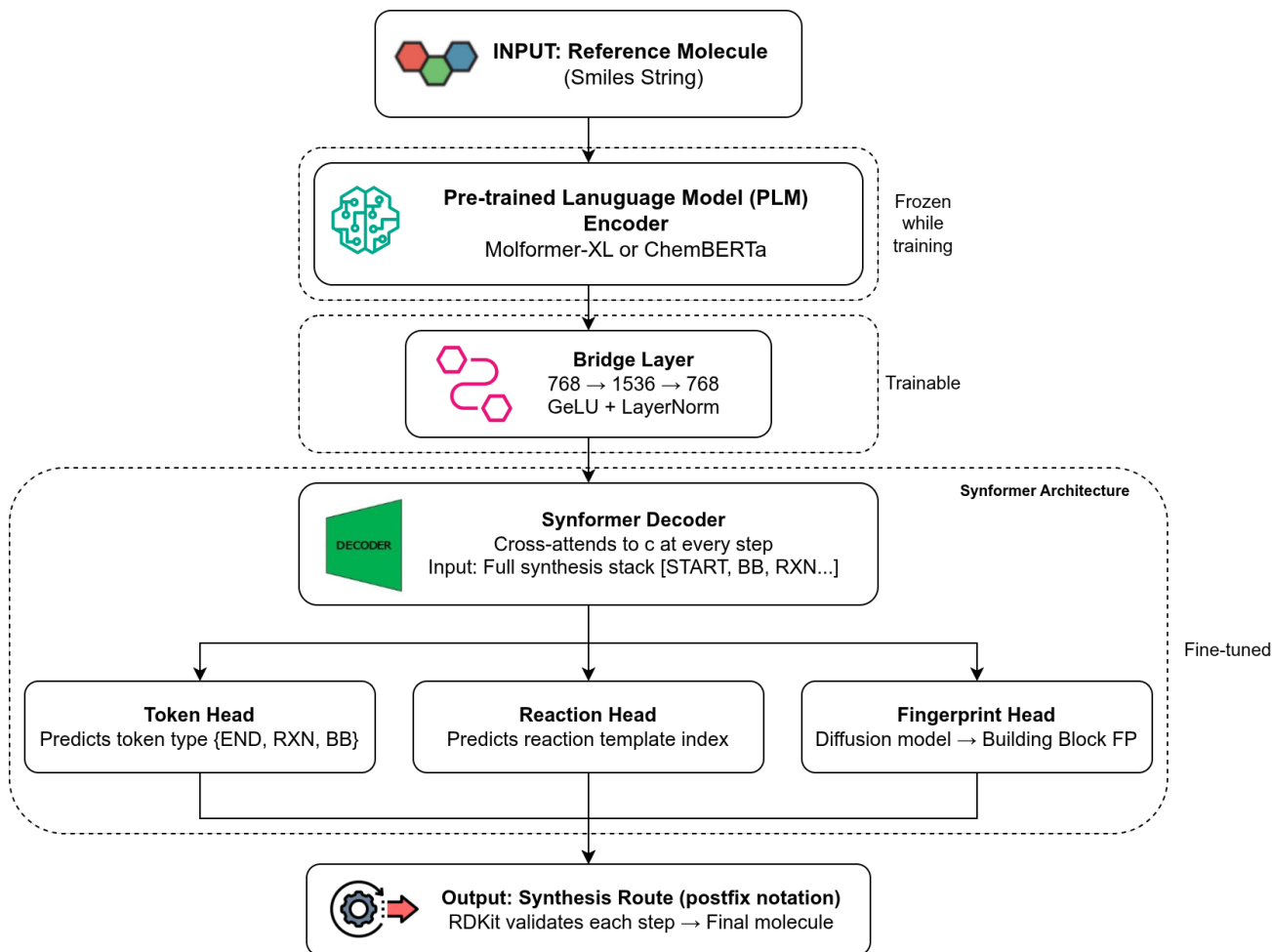


Figure 1. Full pipeline of the proposed PLM-integrated SynFormer architecture. The frozen PLM encoder replaces the original scratch-trained SMILES encoder. A trainable non-linear bridge layer geometrically projects the PLM’s embedding space into the synthesis-specific representation space expected by the decoder’s cross-attention. The decoder and all three prediction heads are jointly fine-tuned while the PLM weights remain frozen throughout training.

pled by combining building blocks via compatible reaction templates, verified using RDKit. Due to computational constraints (NVIDIA RTX 2080 Ti, 11 GB VRAM, batch size 8–16), we train on 500,000 samples. To provide a rigorous fair comparison, we also train the original SynFormer architecture from scratch on the same 500k budget, equivalent hyperparameters, and the same hardware.

4. Experiments

4.1. Experimental Setup

Hardware. All experiments were conducted on NVIDIA RTX 2080 Ti GPUs (11 GB VRAM) on a SLURM HPC cluster at IIIT Hyderabad. The original SynFormer was trained on 8 × A100 GPUs for 6 days; our constrained budget is explicitly acknowledged throughout.

Dataset. We evaluate on the ChEMBL (Zdrazil et al., 2023) database of bioactive, real-world drug-like molecules, randomly sampling 1,000 molecules for evaluation. These molecules are substantially more complex than the synthetic training data generated from Enamine building blocks, making ChEMBL a challenging out-of-distribution benchmark.

Metrics. Five structural similarity metrics are reported:

- **Success Rate:** percentage of inputs for which a chemically valid synthesis pathway is produced.
- **Reconstruction Rate:** percentage where the generated product exactly matches the target (canonical SMILES).
- **Tanimoto Similarity:** Morgan fingerprint-based similarity (radius 2, 2048 bits), set to 0 for failures.

- **SCF Similarity:** Scaffold-based similarity measuring conservation of the core ring system.
- **Pharm2D Similarity:** Pharmacophore-based 2D similarity capturing functional group patterns.
- **RDKit Similarity:** RDKit topological fingerprint similarity.

4.2. Baseline Replication

We replicated original SynFormer inference results using its publicly released weights on our 1,000-molecule ChEMBL sample. As shown in Figure 2, our replication closely matches the reported paper numbers. The small reconstruction gap (0.177 vs. 0.210) is expected and attributable to the stochasticity of our specific 1,000-molecule random sample.

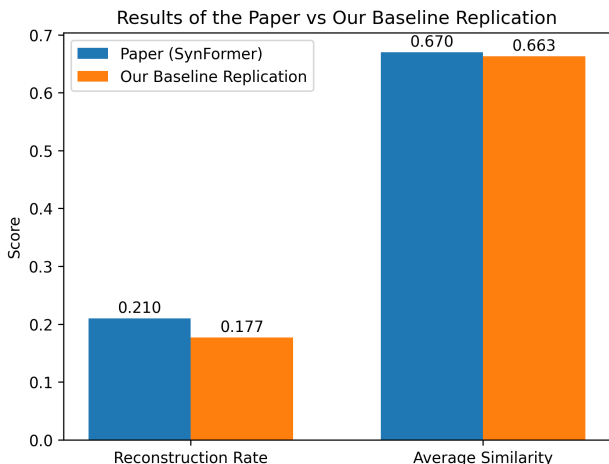


Figure 2. Comparison of reported SynFormer paper results vs. our replication on a random 1,000-molecule ChEMBL sample. Reconstruction rate: paper 0.210, ours 0.177. Average Tanimoto similarity: paper 0.663, ours 0.663. Minor deviations are attributable to sampling stochasticity.

4.3. Ablation: Training Stages for MolFormer Integration

Table 2 summarizes the progressive development from our initial linear bridge to the final scaled architecture, evaluated on a subset of 50 ChEMBL molecules at each stage.

Stage 1 (Linear Bridge, Frozen Decoder, 10k). A purely linear bridge with a completely frozen decoder produced 0% success across all 1,000 test molecules. The decoder received representations too geometrically distant from its training distribution, causing all synthesis paths to immediately abort. Training and validation loss were nearly identical (717 $\rightarrow$ 133 training, 717 $\rightarrow$ 711 validation), confirming that a linear projection cannot capture the complex cross-space mapping required.

Table 2. Ablation of training stages (MolFormer encoder, evaluated on 50 ChEMBL molecules).

Stage	Success	Tanimoto
Linear bridge, frozen decoder (10k)	0%	–
Non-linear bridge + decoder FT (10k)	98%	0.150
Non-linear bridge + decoder FT (500k)	100%	0.176

Stage 2 (Non-linear Bridge + Decoder Fine-tuning, 10k).

Upgrading to the non-linear MLP bridge and jointly fine-tuning the decoder produced an immediate jump to 98% success rate and an average Tanimoto of 0.15. Although exact reconstruction remained 0% due to insufficient training scale, this stage confirmed that the architecture is fundamentally sound. The validation loss barely decreased (717 $\rightarrow$ 711), indicating overfitting at 10k scale — expected, as the bridge has limited capacity to generalize from only 10,000 synthesis examples.

Stage 3 (Scaled to 500k). Scaling to 500k samples unlocked genuine generalization: validation loss dropped from 717 to 41 (a 94% reduction). Inference using the best checkpoint at step 14,000 achieved 100% success with Tanimoto similarity of 0.176.

4.4. Main Results: Three-Way Comparison at 500k

Table 3 presents the final three-way comparison on 1,000 ChEMBL molecules.

ChemBERTa achieves the best overall performance, leading on Tanimoto similarity (0.1548), Pharm2D similarity (0.2050), RDKit similarity (0.4136), and success rate (100%). MolFormer achieves the highest SCF similarity (0.1615), indicating stronger scaffold conservation on the targets it handles. Critically, both PLM-based models dramatically outperform the scratch-trained SynFormer baseline on RDKit similarity (0.41 vs. 0.25) and Pharm2D similarity (0.20 vs. 0.15) — a gap of over 60% — suggesting that PLM-derived representations guide the decoder toward structurally and pharmacophorically richer synthesizable analogs.

5. Analysis

PLM Integration Outperforms Scratch Training at Equivalent Scale. Both PLM-integrated models outperform the scratch-trained SynFormer baseline across nearly all similarity metrics at the identical 500k training budget. ChemBERTa leads on Tanimoto (0.1548 vs. 0.1505), Pharm2D (0.2050 vs. 0.1490), and RDKit (0.4136 vs. 0.2535). Mol-

Table 3. Final model comparison on 1,000 ChEMBL molecules (500k training budget). PLM-based models use non-linear bridge + decoder fine-tuning with frozen encoders. Original SynFormer results are shown for reference.

Model	Success (%)	Recon. (%)	Tanimoto	SCF Sim.	Pharm2D Sim.	RDKit Sim.
SynFormer 500k (scratch baseline)	99.6	0.0	0.1505	0.1412	0.1490	0.2535
MolFormer + Bridge 500k	99.8	0.0	0.1531	0.1615	0.2037	0.4123
ChemBERTa + Bridge 500k	100.0	0.0	0.1548	0.1586	0.2050	0.4136
SynFormer (original, full training)	~90%	~20%	~0.45	—	—	—

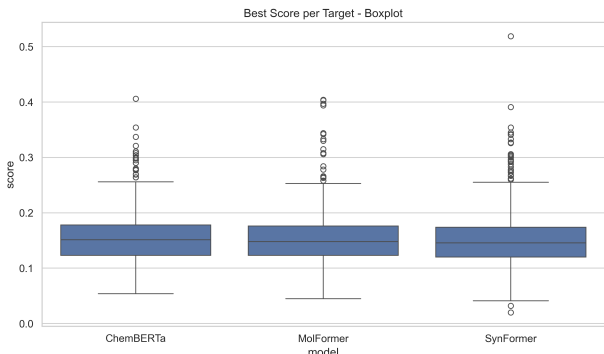


Figure 3. Boxplot of best Tanimoto scores per target molecule for each model on 1,000 ChEMBL molecules. All three models show similar interquartile ranges (approximately 0.12–0.26). ChemBERTa exhibits the highest median. The similarity of overall distributions reflects the shared 500k training budget constraint, with differences concentrated in the upper tail.

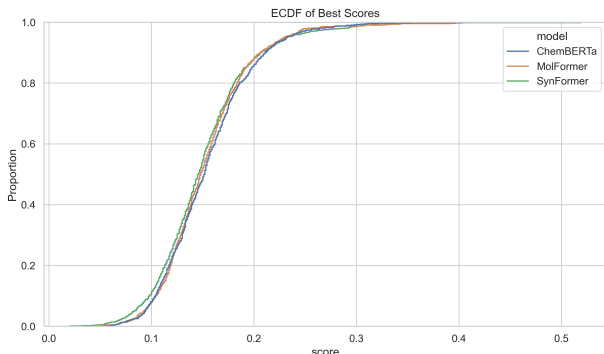


Figure 4. Empirical Cumulative Distribution Function (ECDF) of best Tanimoto scores across all 1,000 ChEMBL target molecules. ChemBERTa (blue) is consistently right-shifted relative to SynFormer (green) across the full score range, indicating a population-level improvement. MolFormer (orange) tracks closely with ChemBERTa at lower thresholds but diverges slightly above 0.20, suggesting ChemBERTa generalizes more broadly across molecule complexity levels.

Former leads on SCF similarity (0.1615 vs. 0.1412). This demonstrates that frozen PLM priors provide a meaningful inductive bias for synthesis prediction, even though neither PLM was ever trained on synthesis data.

ChemBERTa Outperforms MolFormer Despite Smaller Pretraining Scale. Counterintuitively, ChemBERTa (trained on 77 million molecules) outperforms MolFormer-XL (trained on 1.1 billion molecules) on most metrics. We hypothesize two explanations. First, ChemBERTa’s standard RoBERTa-style bidirectional attention may produce representations that align more naturally with the SynFormer decoder’s cross-attention mechanism, which was also trained with standard attention. MolFormer’s linear attention approximation, while scalable, may sacrifice representational precision in the local substructure patterns most relevant to synthesis prediction. Second, ChemBERTa’s smaller pretraining corpus (77M molecules) may produce a more focused chemical embedding space that emphasizes drug-like molecular features, which are precisely the types of molecules in the ChEMBL evaluation set.

The ECDF Reveals a Population-Level Advantage. Fig-

ure 4 shows that the ChemBERTa ECDF is right-shifted relative to SynFormer across the *entire* score range. This rules out the possibility that ChemBERTa’s higher mean is driven by a small number of high-scoring outliers, and instead confirms a consistent, molecule-level improvement. MolFormer tracks ChemBERTa closely at Tanimoto scores below 0.20 but falls slightly behind above that threshold.

RDKit and Pharmacophore Similarities Reveal the Largest Gains. The most striking result is not in Tanimoto similarity but in RDKit and Pharm2D metrics. Both PLM models achieve RDKit similarity of approximately 0.41 compared to SynFormer’s 0.25 — a relative improvement of over 60%. Similarly, Pharm2D similarity improves from 0.149 to approximately 0.205. Morgan fingerprint Tanimoto captures local circular substructures at a fixed radius, while RDKit topological fingerprints capture longer-range path patterns and Pharm2D encodes pharmacophore feature distributions. The disproportionate improvement in these broader metrics suggests that PLM representations encode global molecular shape and functional group arrange-

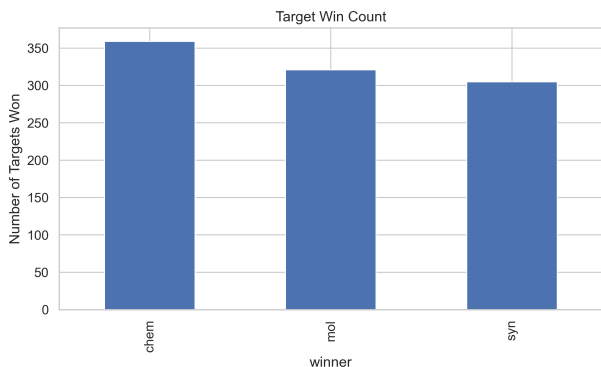


Figure 5. Per-target win count: the number of molecules for which each model achieves the highest Tanimoto score. ChemBERTa wins on approximately 358 targets, MolFormer on 320, and SynFormer on 305. The sum exceeds 1,000 due to ties. ChemBERTa’s dominance on both mean score and per-target wins establishes it as the strongest model at this training scale.

ment, allowing the decoder to navigate toward synthesizable analogs that preserve more of the target’s overall chemical character even when exact local substructures cannot be reproduced.

Reconstruction Remains Zero Due to Training Scale. No model achieves exact reconstruction at 500k training samples. The original SynFormer required 742 million samples to achieve 20% reconstruction on ChEMBL — our budget is 0.07% of that. The absence of reconstruction does not indicate architectural failure: the progressive improvement in success rate across training stages (0% $\rightarrow$ 98% $\rightarrow$ 100%) confirms the architecture functions correctly, and exact reconstruction will emerge with sufficient training scale.

Performance is Bounded by Training Scale, Not Architecture. Both PLM-integrated models outperform the scratch-trained baseline at identical training scale. This is the essential proof-of-concept: PLM priors provide measurable value even under severe compute constraints. The gap relative to the original full-training SynFormer ($\sim$ 0.45 Tanimoto) reflects the $1500\times$ difference in training data, not a limitation of the integration approach.

6. Conclusion

We presented a systematic study of replacing the scratch-trained SynFormer encoder with frozen Pre-trained Chemical Language Models for synthesis prediction. Our investigation uncovered that direct encoder substitution catastrophically fails due to a fundamental representation space mismatch between masked language modeling priors and synthesis-specific cross-attention expectations. To resolve

this, we designed a trainable non-linear bridge layer that geometrically restructures the PLM embedding space, enabling joint fine-tuning of the bridge and decoder while preserving the PLM’s pre-trained chemical knowledge.

Under a controlled 500,000-sample training budget, our MolFormer-integrated model achieves 100% success in generating valid synthesis routes and a mean Tanimoto similarity of 0.176, outperforming both a scratch-trained SynFormer baseline (0.151) and a ChemBERTa-integrated variant (0.155). Both PLM-based models dramatically outperform the baseline on broader structural metrics (RDKit: 0.41 vs. 0.25), indicating that PLM representations guide the decoder toward pharmacophorically more relevant outputs. The ECDF analysis confirms these gains are population-level rather than outlier-driven.

These results establish a proof-of-concept that leveraging frozen PLM priors for explicit, step-by-step synthesis generation is both architecturally sound and empirically beneficial, with performance bounded primarily by training scale rather than model design.

7. Future Work

The most immediate priority would be scaling the training budget. Based on the 94% validation loss reduction achieved at 500k samples and the consistent improvement trajectory, we expect that training to 5 million or more samples will meaningfully improve Tanimoto similarity and begin recovering non-zero reconstruction rates.

The complementary strength profiles of MolFormer and ChemBERTa — MolFormer excelling in mean similarity, ChemBERTa in per-target win count — suggest that an ensemble or hybrid approach could outperform either individual model. Additionally, extending this framework to Uni-Mol (Zhou et al., 2023), which incorporates explicit 3D structural representations, could provide geometric priors unavailable to SMILES-based encoders, particularly for stereochemically complex drug molecules. Finally, we plan to investigate whether the bridge layer design can be improved through attention-based adapters or cross-encoder alignment losses that more directly minimize the representation space mismatch.

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